

Ling Chang

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EDUCATION

Rice University <i>Master in Computational Science and Engineering</i>	Houston, TX <i>Aug. 2025 - Present</i>
The University of Texas at Dallas <i>Master of Science in Information Technology and Management</i>	Dallas, TX <i>Aug. 2024 - Jul. 2025</i>
National ChengChi University <i>Bachelor of Art in International Relations</i>	Taipei, Taiwan <i>Aug. 2013 - Jul. 2017</i>

SKILLS

Programming Languages: Python, SQL, NoSQL, C#, R, Javascript

Applications: MongoDB, PostgreSQL, Hadoop, Spark, Tableau, Shell Script, HTML/CSS, Git, Linux, SPSS

Methodologies: Machine Learning (ML), Data Analytics, A/B Testing, Statistical Hypothesis Testing

SELECTED PROJECTS

DiscoverKG2Prompt: Knowledge Graphs for LLM Prompt Construction

React.js, Flask, Neo4j, Large Language Models (LLMs), Model Context Protocol (MCP) *Jun. 2025*

- Extended a biomedical KG-based retrieval system to support dynamic LLM prompt refinement with LLM agents and search tools deployed on MCP servers.
- Built a Neo4j-backed graph architecture for real-time semantic and entity-based prompt generation.
- Integrated React.js front-end features for intuitive user-guided exploration of the knowledge graph.
- Developed Flask APIs to support responsive KG visualization and KG-guided prompt construction.

Credit Card Approval Modeling and Prediction

R, Logistic Regression, Decision Tree, Shiny *May 2025*

- Cleaned and categorized skewed financial and demographic features to model credit card approval outcomes.
- Trained interpretable ML models (Decision Tree, Logistic Regression) achieving over 90% accuracy, minimizing false approvals.
- Visualized feature importance and model performance using ROC curves, confusion matrices, etc.
- Built an interactive Shiny app to simulate applicant profiles and provide real-time approval predictions for decision support.

Flight Reservation Database System Design

MySQL, PostgreSQL *Dec. 2024*

- Designed an entity relationship diagram covering customers, flights, bookings, payments, and airports.
- Deployed normalized data definition language (DDL) scripts in third normal form (3NF), improving query efficiency by 30% and reducing redundancy by 20%.
- Constructed data manipulation language (DML) queries and executed business analytics using joins, sub-queries, and aggregate functions.

EXPERIENCE

Border Affairs Corps, National Immigration Agency

Senior Immigration Officer *Taipei, Taiwan*
Feb. 2019 - Aug. 2024

- Led an intelligence analysis team, driving a 50% increase in the performance of risk control.
- Analyzed over 1,000 case data entries for offline and online testing, establishing a machine learning system with anomaly detection algorithms to improve criminal identification accuracy by 40%.